

parameter_analysis

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1 Calcium Model Parameter Sensitivity Analysis

Systematic quantification of how measurement noise, temporal filtering, and indicator saturation degrade the fidelity of calcium-imaging recordings. Implements the forward model from Deneux et al. (2016) to study three key parameter interactions:

1. **Noise vs. Smoothing:** Trade-off between stochastic corruption and temporal resolution
2. **Saturation vs. Noise:** Nonlinear distortion vs. measurement uncertainty
3. **Saturation vs. Firing Rate:** Activity-dependent saturation in fast-firing neurons

1.1 Biophysical Model

Following Deneux et al. (Nature Communications, 2016), normalized calcium dynamics with saturating fluorescence indicator:

$$\frac{dc}{dt} = s(t) - \frac{c(t)}{\tau}$$

$$\frac{dB}{dt} = \eta dW(t)$$

$$F(t) = B(t) \left(1 + A \frac{c(t)}{1 + g c(t)} \right) + \sigma \varepsilon(t)$$

where: - $c(t)$ is normalized intracellular calcium concentration (at rest $c = 0$; after single AP $c = 1$) - $s(t) = \sum_i \delta(t - t_i)$ is the spike train (Dirac sum) - τ is the calcium decay time constant (population mean: 0.81 s for OGB-1) - $B(t)$ is drifting baseline fluorescence (Brownian motion if $\eta > 0$) - $F(t)$ is measured fluorescence intensity - A is relative fluorescence increase per spike ($\Delta F/F$; OGB-1 mean: 5.2%) - g is saturation parameter: $g = \frac{\Delta[C]_T}{[Ca]_0 + K_d}$ (inverse spike count at half-saturation; $g = 0$ recovers linear limit) - η is baseline drift amplitude (set to 0 for flat baseline) - σ is white measurement-noise standard deviation - $\varepsilon(t)$ is unit-variance Gaussian noise

Signal fidelity is quantified as Pearson correlation between post-processed calcium trace and ground-truth spike-driven firing rate (deconvolution removes indicator dynamics).

```
[ ]: using DrWatson  
  
using SpikingNeuralNetworks  
using SNNModels
```

```

using CairoMakie
using ProgressBars

@load_units

include("CA_model.jl")

```

1.2 Numerical Implementation

Calcium integration: Exact exponential decay between discrete time bins: $c[i+1] = c[i]e^{-\Delta t/\tau} + n_i$ where n_i is spike count in bin i and $\Delta t = 1/f_s$ (here $f_s = 50$ Hz).

Deconvolution: L2-regularized inverse filtering to recover firing rate from $\Delta F/F$ signals, separately for calcium dynamics and exponential double-exponential model (baseline for comparison).

Quality metric: Kendall correlation (robust to outliers) between deconvolved and ground-truth rates.

1.3 Experimental Design

Standard conditions across all analyses: - Baseline firing rate: 5 Hz (Poisson process) - Baseline saturation: $g = 0.01$ (modest nonlinearity) - Baseline noise: $= 0$ (noise-free, varied per analysis) - Baseline smoothing: `_smooth = 100 ms` (temporal filtering postprocessing) - Monte Carlo: 10 independent Poisson spike trains per condition - Duration: 50 seconds per realization

Each analysis sweeps two parameters orthogonally while holding the rest fixed.

```

[ ]: # Fixed parameters
n_realizations = 10
simulation_time = 50s
dt = 20ms

# Base biophysical model parameters
base_params = CaModel(
    = 2s,
    A = 0.2,
    F0 = 1.0,
    = 0.0,
)

```

1.4 Comparison: Biophysical vs. Phenomenological Models

The following visualization compares three signal recovery pipelines applied to simulated calcium recordings:

1. Biophysical model (MLspike): Deneux et al. forward model with estimated parameters (τ , A , g) **2. Exponential double-exponential:** Heuristic two-stage convolution (rise + decay), parameter-free **3. Ground truth:** Actual spike-driven firing rate (unobserved, used for benchmarking only)

Conditions across the two models: - Firing rate: 5 Hz - Saturation g : 0.01 - Measurement noise : 0.01 - Smoothing kernel: 100 ms - Realizations: 10 independent Poisson spike trains per condition - Simulation: 50 seconds $\rightarrow$ compute correlations on deconvolved signals

Four-panel layout displays: - **Upper panels:** Raw calcium signals ($\Delta F/F_0$) for both models alongside deconvolved activity traces - **Time series correlation:** Cross-correlation matrices between all signal pairs ($\Delta F/F$ samples) - **Correlation heatmap:** Quantitative comparison (Pearson r) of deconvolved signals vs. ground truth

This cell demonstrates how saturation ($g = 0$) introduces systematic bias in exponential-model deconvolution, while proper biophysical inversion can partially recover spike timing despite nonlinearity.

```
[ ]: include("Ca_convolution.jl")

params = CaModel(
    = 2s,
    A = .2,
    g = 0.01,
    F0 = 1.0,
    = 0.0,
    = 0.1,
)
= 10ms
input_rate = 5Hz
run_comparison(params, , input_rate)
```

```
[ ]: include("Ca_convolution.jl")

params = CaModel(
    = 2s,
    A = .2,
    g = 0.01,
    F0 = 1.0,
    = 0.0,
    = 0.1,
)
= 100ms
input_rate = 5Hz
run_comparison(params, , input_rate)
```

1.5 Analysis 1: Noise vs. Smoothing Trade-off

Fixed parameters: $g = 0.01$ (moderate saturation), firing rate = 15 Hz

Swept parameters: - Noise : 20 log-spaced values from 0.01 to 1.0 - Smoothing kernel `__smooth`: 10 linear values from 1 to 200 ms

Finding: At fixed saturation, noise and smoothing compete. Too little smoothing preserves noise artifacts; too much destroys spike timing information. Optimal smoothing depends on noise level.

1.5.1 Results: Noise vs. Smoothing Heatmaps

Three correlation matrices show how ground truth correlates with calcium model, double-exponential model, and the models' cross-correlation.

```
[ ]:
```

1.5.2 Very low non-linearity

```
[ ]: @unpack calcium, double_exp, cross = noise_smooth_every_g[0.0001]
fig = plot_correlation_heatmaps(
    smooth_range ./ ms,
    _range,
    calcium,
    double_exp,
    cross,
    x_label = "Smoothing (ms)",
    y_label = "Noise ",
    subtitle = "(g = 0.0001, 15Hz)",
    x_scale = Makie.identity,
    y_scale = log10
)
fig
```

```
[ ]: @unpack calcium, double_exp, cross = noise_smooth_every_g[0.001]
fig = plot_correlation_heatmaps(
    smooth_range ./ ms,
    _range,
    calcium,
    double_exp,
    cross,
    x_label = "Smoothing (ms)",
    y_label = "Noise ",
    subtitle = "(g = 0.001, 15Hz)",
    x_scale = Makie.identity,
    y_scale = log10
)
fig
```

1.5.3 Non-linearity comparable to σ

```
[ ]: @unpack calcium, double_exp, cross = noise_smooth_every_g[0.01]
fig = plot_correlation_heatmaps(
    smooth_range ./ ms,
    _range,
    calcium,
    double_exp,
    cross,
```

```

x_label = "Smoothing (ms)",
y_label = "Noise ",
subtitle = "(g = 0.1, 15Hz)",
x_scale = Makie.identity,
y_scale = log10
)
fig

```

1.6 Analysis 2: Saturation vs. Noise Interaction

Fixed parameters: smoothing $_smooth = 50$ ms, firing rate = 5 Hz

Swept parameters: - Saturation g : 20 log-spaced values from 0.0001 to 0.1 - Noise : 20 log-spaced values from 0.01 to 1.0

Finding: Saturation and noise interact nonlinearly. At low saturation ($g < 0.001$), correlation is primarily limited by noise. At high saturation ($g > 0.01$), nonlinear compression dominates even at low noise, causing fundamental signal distortion that postprocessing cannot recover.

```

[ ]:
[ ]: fig_g_sigma = plot_correlation_heatmaps(
    g_range_g_sigma,
    _range_g_sigma,
    corr_calcium_g_sigma,
    corr_doubleexp_g_sigma,
    corr_cross_g_sigma,
    x_label = "Saturation g",
    y_label = "Noise ",
    subtitle = "(smoothing = 50ms, rate = 15Hz)"
)
fig_g_sigma

```

1.7 Analysis 3: Saturation $\times$ Firing Rate Activity Dependence

Fixed parameters: smoothing $_smooth = 50$ ms, noise = $10^{-1.5}$

Swept parameters: - Saturation g : 20 log-spaced values from 0.01 to 1.0 - Firing rate: 20 log-spaced values from 0.1 to 20 Hz

Finding: Saturation effects are strongly activity-dependent. At low firing rates (< 1 Hz), calcium remains mostly in the linear regime regardless of g . At high firing rates (> 10 Hz), rapid calcium accumulation pushes even moderately saturating indicators ($g=0.1$) deep into the nonlinear saturation regime, causing severe signal distortion.

```

[ ]:
[ ]: fig_g_rate = plot_correlation_heatmaps(
    g_range_g_rate,
    firing_rate_range_g_rate,

```

```

    corr_calcium_g_rate,
    corr_doubleexp_g_rate,
    corr_cross_g_rate,
    x_label = "Saturation g",
    y_label = "Firing Rate (Hz)",
    subtitle = "(smoothing = 50ms,  $\tau = 10^{-1} \cdot$  )"
)
fig_g_rate

```

1.8 Summary: Three Dimensions of Signal Degradation

This notebook systematically characterizes how different factors undermine calcium imaging fidelity:

Analysis 1 — Noise vs. Smoothing: At a fixed saturation level ($g=0.01$), measurement noise and temporal smoothing present a fundamental trade-off. Small amounts of smoothing cannot remove noise without destroying spike timing; excessive smoothing blurs fast dynamics.

Analysis 2 — Saturation vs. Noise: Saturation and noise are dependent sources of degradation. At low saturation, noise is less detrimental.

Analysis 3 — Saturation vs. Firing Rate: The practical impact of saturation critically depends on neuronal activity. Sparse neurons (<1 Hz) remain in the linear regime; fast-firing neurons (>10 Hz) accumulate calcium rapidly, pushing even moderately saturating indicators into nonlinearity.